Basic Commands

```sql
-- Create a database
CREATE DATABASE EMPLOYEE;
```

Logs:

```
Executing: CREATE DATABASE EMPLOYEE
Result: 1 rows affected in 13ms
```

```sql
-- Use a database
USE EMPLOYEE;
```

Logs:

```
Executing: USE EMPLOYEE
Result: 0 rows affected in 5ms
```

```sql
-- Drop a database
DROP DATABASE Employee;
```

Logs:

```
Executing: DROP DATABASE Employee
Result: 0 rows affected in 32ms
```

Danishwer
0026MCA25

```sql
-- Create a table in a database
CREATE TABLE students (
    roll_no INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(50),
    city VARCHAR(50),
    marks FLOAT
);
-- Display all tables in a database
SHOW TABLES;
```

Tables_in_company_db varchar
students

```sql
-- Describe a table : Used to display the structure of a table
DESCRIBE students;
```

Field varchar	Type blob
roll_no	int
name	varchar(50)
department	varchar(50)
city	varchar(50)
marks	float

```sql
--Drop: Used to delete a table
DROP TABLE students;


-- Turncate: Used to empty a table
TRUNCATE TABLE students;



-- Alter

--Adding a new column
ALTER TABLE students ADD COLUMN phone VARCHAR(20);
```

1 Total 0

roll_no int	name varchar(50)	department varchar(50)	city varchar(50)	marks float	phone varchar(20)

```sql
-- Modify a column
ALTER TABLE students MODIFY COLUMN phone VARCHAR(10);
```

1 Total 0

roll_no int	name varchar(50)	department varchar(50)	city varchar(50)	marks float	phone varchar(10)

```sql
-- rename a column
ALTER TABLE students RENAME COLUMN phone TO phone_no;
```

roll_nc int	name varchar(50)	department varchar(50)	city varchar(50)	marks float	phone_no varchar(10)

```sql
-- Drop a column
ALTER TABLE students DROP COLUMN phone_no;
```

roll_nc int	name varchar(50)	department varchar(50)	city varchar(50)	marks float

Danishwer

0026MCA25

Practical

Student Table:

```
-- Select Data
SELECT * FROM students;
```

roll_no int	name varchar(50)	department varchar(50)	city varchar(50)	marks float
1	Aditi	CS	Delhi	85
2	Manav	IT	Mumbai	76
3	Riya	CS	Delhi	92
4	Kabir	Math	Pune	88
5	Sneha	IT	Mumbai	64
6	Arjun	Math	Delhi	91

INDEX

S.no.	Title
1	Write a query to count students based on city
2	Write a query to find the max marks for each department
3	Write a query to find the departments having average marks greater than 80.
4	Write a query to delete all students of the IT department.
5	Write a query to rename a column
6	Write a query to show average marks for each department

Danishwer

0026MCA25

1. Write a query to count students based on city

```
SELECT city, count(*) as student_count FROM students GROUP BY city;
```

city varchar(50)	student_count
Delhi	3
Mumbai	2
Pune	1

2. Write a query to find the max marks for each department

```
SELECT department, max(marks) as max_marks

FROM students

GROUP BY

    department;
```

department varchar(50)	max_marks
CS	92
IT	76
Math	91

Danishwer

0026MCA25

3. Write a query to find the departments having average marks greater than 80.

```
Select department, avg(marks) as avg_marks
FROM students
GROUP BY
    department
HAVING
    AVG(marks) > 80;
```

department varchar(50)	avg_marks
CS	88.5
Math	89.5

4. Write a query to delete all students of the IT department.

```
DELETE FROM students WHERE department = 'IT';
```

5. Write a query to rename a column

```
Alter TABLE students RENAME COLUMN name TO s_name;
```

roll_nc int	s_name varchar(50)	department varchar(50)	city varchar(50)	marks float

Danishwer

0026MCA25

6. Write a query to show average marks for each department

```sql
Select department, avg(marks) as avg_marks

FROM students

GROUP BY

    department;
```

department varchar(50)	avg_marks
CS	88.5
IT	70
Math	89.5

Danishwer
0026MCA25